Bonding and its impact on properties of materials

Name: ____________________

Class: ____________________

Date: ____________________

Time: **62 minutes**

Marks: **59 marks**

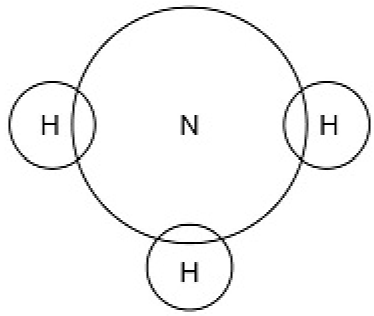
Comments:

1. This question is about ammonia, NH_3

(a) Complete the dot and cross diagram for the ammonia molecule shown in **Figure 1**.

Show only the electrons in the outer shell of each atom.

Figure 1



(2)

(b) Give **one** limitation of using a dot and cross diagram to represent an ammonia molecule.

____

____

(1)

(c) Explain why ammonia has a low boiling point.

You should refer to structure and bonding in your answer.

____

____

____

____

____

____

(3)

Ammonia reacts with oxygen in the presence of a metal oxide catalyst to produce nitrogen and water.

(d) Which metal oxide is most likely to be a catalyst for this reaction?

Tick (✓) **one** box.

CaO ☐

Cr_2O_3 ☐

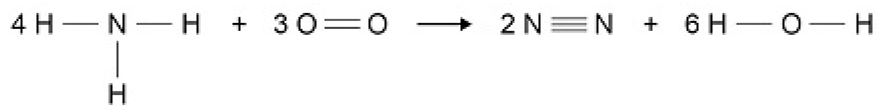
MgO ☐

Na_2O ☐

(1)

Figure 2 shows the displayed formula equation for the reaction.

Figure 2



The table shows some bond energies.

Bond	N — H	O ═ O	N ≡ N	O — H
Bond energy in kJ/mol	391	498	945	464

(e) Calculate the overall energy change for the reaction.

Use **Figure 2** and the table.

Overall energy change = ______________________ kJ/mol

(3)

(f) Explain why the reaction between ammonia and oxygen is exothermic.

Use values from your calculation in part (e).

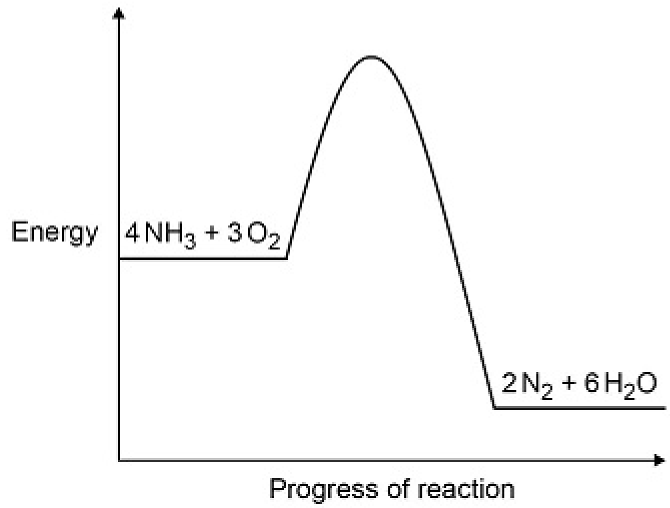
(2)

(g) **Figure 3** shows the reaction profile for the reaction between ammonia and oxygen.

Complete **Figure 3** by labelling the:

- activation energy
- overall energy change.

Figure 3



(2)

(Total 14 marks)

2. This question is about Group 7 elements.

Chlorine is more reactive than iodine.

(a) Name the products formed when chlorine solution reacts with potassium iodide solution.

___

___

(1)

(b) Explain why chlorine is more reactive than iodine.

___

___

___

___

___

___

(3)

(c) Chlorine reacts with hydrogen to form hydrogen chloride.

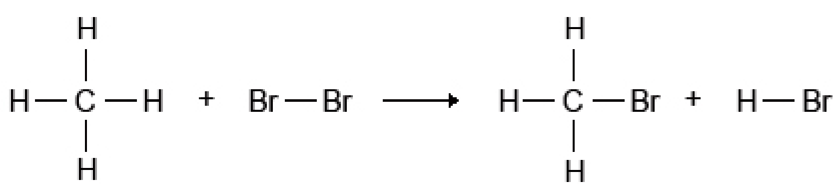
Explain why hydrogen chloride is a gas at room temperature.

Answer in terms of structure and bonding.

(3)

(d) Bromine reacts with methane in sunlight.

The diagram below shows the displayed formulae for the reaction of bromine with methane.



The table below shows the bond energies and the overall energy change in the reaction.

	C — H	**Br—Br**	**C—Br**	**H—Br**	**Overall energy change**
Energy in kJ/mol	412	193	**X**	366	−51

Calculate the bond energy **X** for the C—Br bond.

Use the diagram and the table above.

____

____

____

____

____

____

____

Bond energy **X** = ________________ kJ/mol

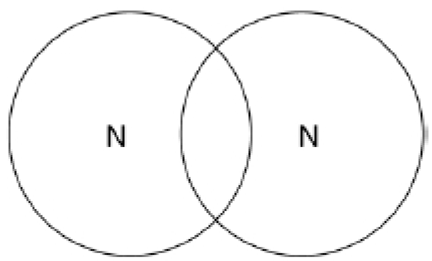
(4)

(Total 11 marks)

3. This question is about structure and bonding.

(a) Complete the dot and cross diagram to show the covalent bonding in a nitrogen molecule, N_2

Show only the electrons in the outer shell.



(2)

(b) Explain why nitrogen is a gas at room temperature.

Answer in terms of nitrogen's structure.

____

____

____

____

____

____

(3)

(c) Graphite and fullerenes are forms of carbon.

Graphite is soft and is a good conductor of electricity.

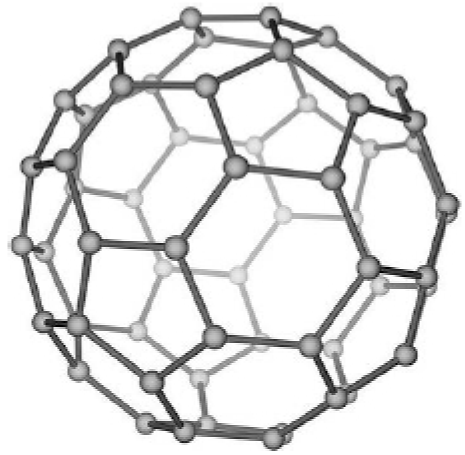
Explain why graphite has these properties.

Answer in terms of structure and bonding.

(4)

(d) **Figure 1** shows a model of a Buckminsterfullerene molecule.

Figure 1



A lubricant is a substance that allows materials to move over each other easily.

Suggest why Buckminsterfullerene is a good lubricant.

Use **Figure 1**.

____

____

____

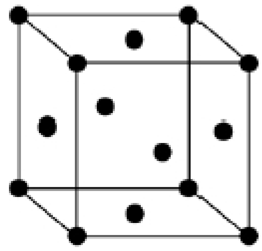
____

(2)

Silver can form cubic nanocrystals.

Figure 2 represents a silver nanocrystal.

Figure 2



(e) A silver nanocrystal is a cube of side 20 nm

Calculate the surface area to volume ratio of the nanocrystal.

____

____

____

____

____

Surface area to volume ratio = ____________________

(3)

(f) Silver nanoparticles are sometimes used in socks to prevent foot odour.

Suggest why it is cheaper to use nanoparticles of silver rather than coarse particles of silver.

____

____

____

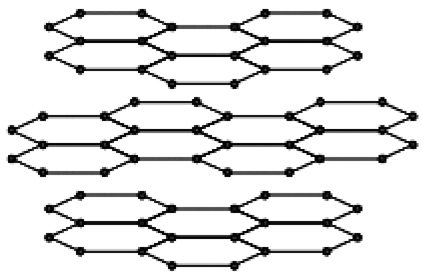
____

(2)

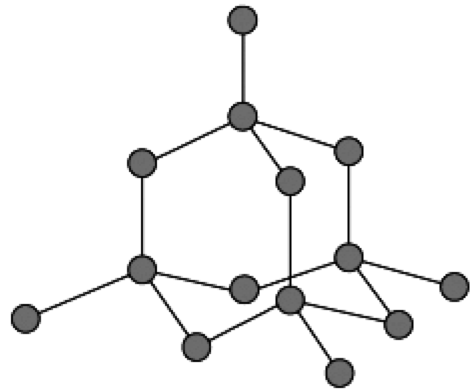
(Total 16 marks)

4. Graphite and diamond are different forms of the element carbon.
Graphite and diamond have different properties.

The structures of graphite and diamond are shown below.



Graphite



Diamond

(a) Graphite is softer than diamond.

Explain why.

____

____

____

____

____

____

____

____

(4)

(b) Graphite conducts electricity, but diamond does not.

Explain why.

____

____

____

____

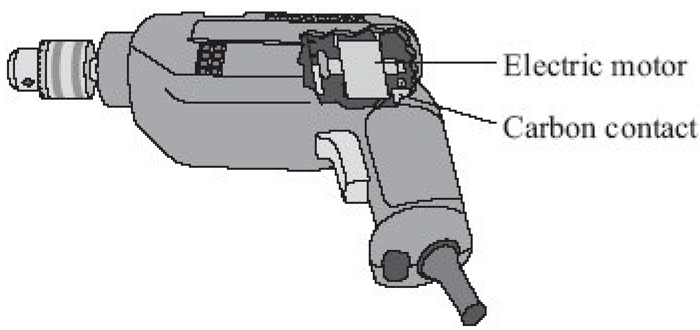
____

____

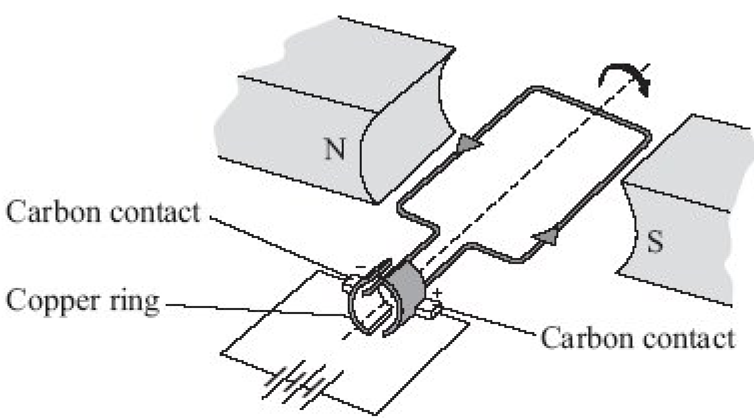
(3)

(Total 7 marks)

5. This drill contains an electric motor.



The diagram below shows the main parts of an electric motor.



The carbon contacts are made of graphite. Springs push the contacts against the copper ring. The contacts conduct electricity to the copper ring. The copper ring rotates rapidly but does not stick or become worn because the graphite is soft and slippery.

Graphite has properties which are ideal for making the contacts in an electric motor.

Explain, in terms of structure and bonding, why graphite has these properties.

(Total 5 marks)

6. This question is about the properties and uses of materials.

Use your knowledge of structure and bonding to answer the questions.

(a) Explain how copper conducts electricity.

(2)

(b) Explain why diamond is hard.

____

____

____

____

____

(2)

(c) Explain why thermosetting polymers are better than thermosoftening polymers for saucepan handles.

____

____

____

____

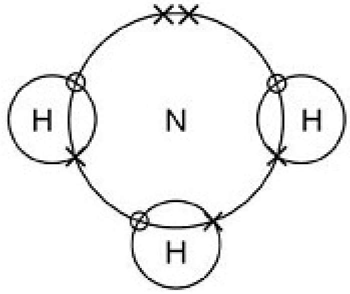
____

(2)

(Total 6 marks)

Mark schemes

1. (a)



*scores **2** marks*

allow dots, crosses, circles or $e^{(-)}$ for electrons

1 bonding pair of electrons in each overlap

1

2 non-bonding electrons on nitrogen

*do **not** accept non-bonding electrons on hydrogen*

ignore inner shell electrons drawn on nitrogen

1

(b) does not show the shape

or

only two-dimensional

allow is not three-dimensional

1

(c) (ammonia has) small molecules

allow (ammonia has) a simple molecular (structure)

1

(ammonia has) weak intermolecular forces

allow (ammonia has) weak intermolecular bonds

*do **not** accept weak covalent bonds*

1

(so) little energy is needed to overcome the intermolecular forces

allow (so) little energy is needed to break the intermolecular bonds

allow (so) little energy is needed to separate the molecules

*do **not** accept references to breaking covalent bonds*

1

(d) Cr_2O_3

1

(e)

*an answer of (-)1272 (kJ) scores **3** marks*

(for bonds broken)
((12 x 391) + (3 x 498) =) 6186

1

(for bonds made)
((2 x 945) + (12 x 464) =) 7458

1

(overall energy change = 6186-7458 =) (-)1272 (kJ)

allow correct calculation using incorrectly calculated values from step 1 and/or step 2

1

(f)

allow ecf from part (e)

7458 (kJ) (released in making bonds) is greater than 6186 (kJ) (used in breaking bonds)
or
the products have 1272 (kJ) less energy than the reactants

allow the (overall) energy change is -1272 (kJ)

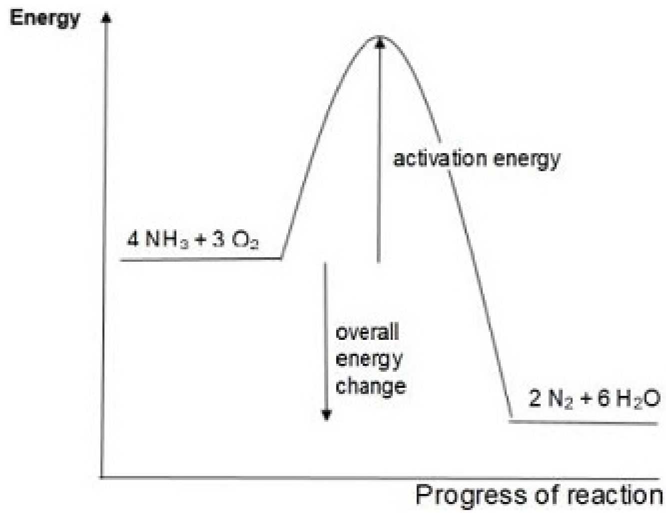
1

(so) energy is released (to the surroundings)

dependent on MP1 being awarded
allow (so) heat is released (to the surroundings)
*if no values given, allow **1** mark for more energy released in making bonds than used in breaking bonds*

1

(g)



*scores **2** marks*

allow discontinuous lines
ignore arrow heads

activation energy labelled

1

(overall) energy change labelled

1

[14]

2. (a) potassium chloride **and** iodine

either order

allow KCl for potassium chloride and I_2 for iodine

1

(b) (chlorine's) outer electrons / shell closer to the nucleus

allow chlorine has fewer shells

allow chlorine atom is smaller than iodine atom

ignore chlorine has fewer outer shells

1

(so) the chlorine nucleus has greater attraction for outer electrons / shell

allow chlorine has less shielding

*do **not** accept incorrect types of attraction*

1

(so) chlorine gains an electron more easily

1

***max 2** marks can be awarded if the answer refers to chloride / iodide instead of chlorine / iodine*

allow converse statements

allow energy levels for shells throughout

(c) hydrogen chloride is made of small molecules

allow hydrogen chloride is simple molecular

1

(so hydrogen chloride) has weak intermolecular forces*

1

(intermolecular forces) require little energy to overcome*

1

do **not accept reference to bonds breaking unless applied to intermolecular bonds*

(d) (bonds broken = 4(412) + 193 =)1841

1

(bonds formed = 3(412) + 366 + **X** =) 1602 + **X**

1

−51 = 1841 − (1602 + **X**)

allow use of incorrectly calculated values of bonds broken and / or bonds formed from steps 1 and 2 for steps 3 and 4

1

(**X** =) 290 (kJ/mol)

allow a correctly calculated answer from use of −51 = bonds formed − bonds broken

1

OR

alternative method ignoring the 3 unchanged C−H bonds

(412 + 193 =) 605 (1)

366 + **X** (1)

−51 = 605 − (366 + **X**) (1)

(**X** =) 290 (kJ/mol) (1)

*an answer of 290 (kJ/mol) scores **4** marks*

*an answer of 188 (kJ/mol) scores **3** marks*

*an incorrect answer for one step does **not** prevent allocation of marks for subsequent steps*

[11]

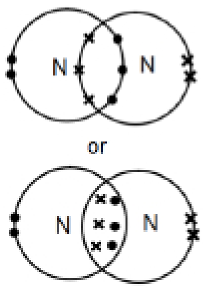
3. (a) six electrons in the overlap

allow dots, crosses or e(-) for electrons

1

2 non-bonding electrons on each nitrogen atom

***2** marks for an answer of:*



1

(b) weak forces

1

between molecules
or
intermolecular

do not allow references to covalent bonding between molecules

1

(which) need little energy to overcome

1

(c) each (carbon) atom forms three covalent bonds

1

forming layers (of hexagonal rings)

1

(soft)
(because) layers can slide over each other

1

(conducts electricity)
(because of) delocalised electrons

1

(d) molecules are spherical

1

(so molecules) will roll

1

(e) surface area (= 20 × 20 × 6) = 2400 (nm^2)

1

volume (= 20^3) = 8000 (nm^3)

1

ratio = 0.3 (nm^3): 1 (nm^3)
ratio = 0.3 (nm^3): 1 (nm^3)
or
1 (nm^3): 3.33 (nm^3)

1

(f) (nanoparticles) have a larger surface area to volume ratio

1

so less can be used for the same effect

1

[16]

4. (a) **Graphite:**

because the layers (of carbon atoms) in graphite can move / slide

it = graphite

1

this is because there are only weak intermolecular forces **or** weak forces between layers

accept Van der Waals' forces allow no <u>covalent</u> bonds between layers

1

Diamond:

however, in diamond, each carbon atom is (strongly / covalently) bonded to 4 others

allow diamond has three dimensional / tetrahedral structure

1

so no carbon / atoms able to move / slide

*allow so no layers to slide **or** so diamond is rigid*

1

(b) because graphite has delocalised electrons / sea of electrons

allow free / mobile / roaming electrons

1

which can carry charge / current **or** move <u>through the structure</u>

1

however, diamond has no delocalised electrons

accept however, diamond has all (outer) electrons used in bonding

1

[7]

5. **five** ideas from the following for one mark each

- each carbon / atom joined / bonded to three other carbon / atoms
  or each carbon forms 3 bonds
- in layers
- only weak forces (of attraction) / bonds between layers

  allow weak electrostatic / intermolecular forces /bonds between layers
- layers / atoms can slide over each other
- one electron on each carbon is not used for bonding
- electrons delocalised **or** electrons free

  allow 'sea' of electrons
- electrons carry the charge / current
- giant structure / lattice
- covalent (bonds)
- strong bonds **or** a lot of energy needed to break bonds

  reference to ionic bonding = ***max 4***

  diagrams could be used:
  - *to show layered structure*
  - *to show that each carbon is bonded to three other carbon atoms*
  - *to show giant structure (at least 3 rings required)*

[5]

6. (a) has delocalised electrons

accept free (moving) electrons

1

(so electrons) can move through the structure/metal

accept (so electrons) can carry charge through the structure/metal

accept (so electrons) can form a current

1

reference to incorrect particles ***or*** *incorrect bonding* ***or*** *incorrect structure =* ***max 1***

(b) giant structure

accept lattice

accept each atom forms four bonds (with other carbon atoms)

ignore macromolecular

1

strong bonds

accept covalent

*do **not** accept ionic*

1

*reference to intermolecular forces/bonds **or** incorrect particles = **max 1***

(c) thermosetting polymers do not melt (when heated)

accept thermosetting polymers do not change shape (when heated)

accept thermosetting polymers have high(er) melting points

ignore thermosetting polymers do not soften (when heated)

1

due to cross-links (between chains)

accept due to bonds between chains

1

*reference to smart polymers = **max 1***

accept converse argument

[6]

Examiner reports

1. (a) Approximately three-quarters of students were able to complete the dot and cross diagram correctly.

The mark for the bonding electrons was scored more often than the mark for the non-bonding electrons.

(b) Only a small proportion of students were able to give a limitation of a dot and cross diagram.

The most common correct response was that the molecule was not shown in 3 dimensions.

Common responses seen that were not credited included:

- only shows the outer electrons
- does not show the structure
- does not show the intermolecular forces.

(c) Students found this question difficult.

The mark for indicating the presence of weak intermolecular forces was most often awarded. However, some students referred to ‘intermolecular forces’ incorrectly, for example, weak intermolecular forces between the atoms / bonds.

A common misconception was that covalent bonds are weak and that it is these bonds that are broken when ammonia boils.

Ammonia consisting of small molecules was the mark least often awarded.

(d) Students found identification of a metal oxide most likely to be used as a catalyst difficult.

The correct answer, the transition metal oxide, Cr_2O_3, was the most common choice of catalyst but the other 3 options were all chosen in significant numbers, especially MgO.

(e) Approximately two-thirds of students correctly calculated the overall energy change.

The most common error was to incorrectly determine, or not determine, the number of each type of bond from the given equation.

(f) Students found this question challenging. In particular, students did not always use values from their previous calculation as instructed in the question. A common non-scoring response was simply to define an exothermic reaction.

Few students compared their calculated values for bond breaking and bond making as was intended. Some referred in general terms to these processes but often used incorrect terms, for example, more energy was <u>needed</u> for making bonds than used for breaking bonds.

(g) The completion of the energy profile was often done well with almost two-thirds of students scoring both marks.

Some students were imprecise with the positioning of their lines.

Other common errors included:

- Not drawing a line for activation energy, just labelling the peak.
- Drawing a line from the peak to the products line for ‘overall energy change’.

2.

(a) This question was answered well by the majority of students, with 68% giving a correct answer. The most common error was to give only one of the products, usually potassium chloride.

(b) Although this historically is a well-known area of the specification, lack of precision in answers cost some students greatly. 16% of students achieved all three marks, with 72% gaining some credit.

The electrons are not (all) closer to the nucleus in chlorine − the **outer** electrons are. The relevant force of attraction is between the **outer** electrons and the **nucleus**, and is not intermolecular, magnetic or gravitational. Attracting another electron is not the same as **gaining** it.

(c) This was another difficult question with 51% of students gaining no marks at all. 9% of students achieved all three. Few described the structure correctly. ‘Simple covalent’ only refers to the bonding; the structure needed to be described as small molecules, as in the specification, or as simple molecular (to distinguish it from large molecules or giant structures).

Some answers started well, referring to weak intermolecular forces, but then moved into talking about breaking bonds, for example ‘it has weak intermolecular forces so little energy is needed to break the bonds’. Again, language was an issue, with statements such as ‘hydrogen chloride is a covalent bond’ failing to gain any credit. Some students stated that hydrogen chloride is ionic (but often then referred to intermolecular forces), or did not refer to hydrogen chloride at all, instead couching their answer in terms of hydrogen and chlorine.

(d) This question was answered well, with 73% students achieving three or four marks.

It was often difficult to give partial credit as working was difficult to follow. Many students worked out the sum of the bonds broken as 1841 kJ/mol, but then said that the sum of the bonds made was 1602 kJ/mol rather than (1602 + X) kJ/mol. Many students then calculated their final answer based on energy change = bonds made − bonds broken, rather than the other way round.

Some students missed out on putting brackets around their (1602 + X) in their overall expression, making it incorrect. A decent number divided the bond enthalpies by 2 then subtracted one from another to get the difference, which they then added to another halved bond energy. This perhaps shows a misconception that the bond energy is associated with an atom rather than a bond.

4. (a) This question proved to be a good discriminator. The marking point relating to the intermolecular forces in graphite was gained by only the most able students, but that relating to layers sliding in graphite proved the most accessible. A small minority of students referred to incorrect bonding (eg intermolecular forces in diamond) or structure (simple molecular) in their answers. A small minority of the students also referred to crosslinking and thermosetting. Some students mistook softness for flexibility, referring to graphite being able to bend. Some students were content to explain why graphite was soft but then failed to go on to address why diamond was not soft, therefore not gaining the marks for that part of the mark scheme.

(b) Conductivity in graphite and diamond was well understood by many candidatesstudents, although some failed to refer to the delocalised electrons as charge or current carriers or as moving throughout the structure so did not gain the second marking point. A small minority of students were unable to relate conductivity to electron structure at all, and some referred to the need for graphite to be molten to conduct electricity.

5. There were many excellent answers to this question, which showed that some of the candidates have a deep understanding of the structure and bonding in graphite. The question did, however, discriminate very well between the candidates, since there was an even spread of marks from five to zero. Most candidates realised that the graphite structure is layered and that the layers can slide over each other. More candidates than in previous years knew that each carbon is bonded to three other carbon atoms and that each atom has one delocalised electron. Many of the candidates realised the significance of the giant covalent structure to the high melting point of graphite.

The question also showed that some candidates have misconceptions regarding the structure and bonding in graphite. A significant number of candidates thought that graphite is a metal or that graphite is made of ions. Some gave confused answers in which they described the structure of diamond rather than graphite. Some explained that it is slippery because it has weak covalent bonds or because it is malleable.

6. (a) A large majority of responses gained at least one mark, usually for the recognition that delocalised or free electrons have a role in electrical conductivity. A commonly seen incorrect response suggested that particles vibrating or colliding passed an electrical charge through the metal. It would appear that these students are confusing electrical conduction with conduction of heat.

(b) Some students were able to clearly explain the relevant structural features and bonding in diamond. Other students found it difficult to express that each atom forms four bonds, although this was a popular response. Some remembered the terms “lattice” or “giant structure”. Many students were limited to a maximum of one mark by referring to intermolecular forces, often as an unfortunate afterthought, sometimes describing them as “strong” and sometimes as “weak”.

(c) Many students gained only the first marking point because they did not explain why thermosetting polymers do not melt when heated. Some students suggested that thermosetting polymers would not conduct heat so well, so would be safer to use as saucepan handles. A number of students referred to the cross-links in thermosetting polymers as strong intermolecular forces. They are covalent bonds, not intermolecular forces which limited them to a maximum of one mark.